

Payment Exception & Reconciliation Workflow Redesign

1. Purpose and Scope

This document specifies the matching logic, exception-aging rules, and audit-logging requirements for the rules-based reconciliation engine described in the BRD. It is written to a level of detail sufficient for an engineering team to implement the matching logic in SQL against three source tables — `internal_ledger`, `payment_gateway_feed`, and `clearing_house_file` — joined on a common `transaction_id`. No such query is executed as part of this project; this document is a specification artifact only.

2. Functional Requirements

FR-001 — Exact Match Rule

A transaction is classified as an Exact Match when `transaction_id`, amount, and currency are identical across the internal ledger and the clearing-house file. Exact Matches are auto-resolved with no manual review and logged with reason code `AUTO_EXACT`.

Specification note for engineering: this corresponds to an `INNER JOIN` on `transaction_id` between `internal_ledger` and `clearing_house_file`, filtered `WHERE internal_ledger.amount = clearing_house_file.amount AND internal_ledger.currency = clearing_house_file.currency`.

FR-002 — Tolerance Match Rule

A transaction is classified as a Tolerance Match when `transaction_id` and currency match, but amount differs by less than the tolerance baseline, defined as **0.05% of the transaction amount**. Tolerance Matches are auto-resolved and routed to a designated FX/Fee variance ledger, logged with reason code `AUTO_TOLERANCE`.

Design rationale (stated explicitly, per project integrity rule): the 0.05% baseline is a pragmatic starting parameter proposed to absorb routine FX rounding and settlement-timing variance between systems. It is not derived from any regulatory standard, is not claimed to reflect an industry benchmark, and is intended to be revisited and tuned once real transaction variance data is available — which, in this portfolio project, never happens. This value should be treated as the project's equivalent of Project A's FR-003 placeholder-pending-validation, except that no later stage in this project validates it against real data, so it remains a stated assumption throughout.

Specification note for engineering: `WHERE internal_ledger.transaction_id = clearing_house_file.transaction_id AND internal_ledger.currency = clearing_house_file.currency AND ABS(internal_ledger.amount - clearing_house_file.amount) / internal_ledger.amount < 0.0005`.

FR-003 — Exception Classification

A transaction that satisfies neither FR-001 nor FR-002 — including any transaction with no matching `transaction_id` found in the counterpart file within a 24-hour ingestion window — is

classified as an Exception and is never auto-resolved. Exceptions are logged with reason code EXCEPTION_OPEN and enter the aging-bucket workflow (FR-004).

Specification note for engineering: this corresponds to a FULL OUTER JOIN between internal_ledger and clearing_house_file on transaction_id, retaining all rows where either side is NULL or where the row fails both the FR-001 and FR-002 filter conditions.

FR-004 — Aging-Based Risk Bucketing

Every open Exception is assigned an aging bucket based on elapsed time since the exception was first logged:

- **0–24 hours:** Low priority — queued for standard operator review
- **24–72 hours:** Medium priority — requires active investigation with a documented status note
- **>72 hours:** High priority — immediate escalation to the Operations Lead and visibility to Finance and Compliance

An exception's bucket is recalculated continuously (not fixed at creation time); an exception that ages past a threshold automatically moves to the next bucket without requiring manual reclassification.

FR-005 — Mandatory Audit Logging (all resolution paths)

Every resolution — whether FR-001 (auto, exact), FR-002 (auto, tolerance), or a manual resolution following FR-003/FR-004 — must be logged with: resolution_timestamp, resolved_by (system identifier for automated resolutions, operator ID for manual ones), reason_code, and resolution_note (free text, mandatory for manual resolutions, auto-populated for automated ones). No resolution path is exempt from logging, including auto-resolved Exact Matches.

This requirement directly satisfies Devraj's (Compliance) Stage 1 concern: no resolution, however trivial, escapes the audit trail.

FR-006 — Manual Resolution Requires Reason Code Selection

When an operator manually resolves an Exception, the system must require selection from a defined, closed list of reason codes (e.g., TIMING_DIFFERENCE, DUPLICATE_ENTRY, MISSING_REFERENCE, CURRENCY_CONVERSION_VARIANCE, GENUINE_ERROR, OTHER — note required). Free-text-only resolution, with no structured reason code, is not permitted.

Design rationale: this prevents the As-Is state's core weakness — inconsistent, operator-dependent resolution rationale that wouldn't survive audit sampling.

FR-007 — High-Priority Bucket Escalation Notification

Any Exception entering the >72-hour bucket (FR-004) must trigger a notification to the Operations Lead and must appear on a Finance-visible aging summary. This requirement does

not specify dashboard implementation (explicitly out of scope per BRD Section 6) — only the underlying rule that such visibility must exist.

FR-008 — Total Value at Risk Tracking

The system must be able to calculate, at any point in time, the sum of unresolved Exception amounts, segmented by aging bucket. This is specified as a query requirement only; no dashboard or report is built in this phase.

Specification note for engineering: SELECT aging_bucket, SUM(amount) AS total_value_at_risk FROM exceptions WHERE status = 'OPEN' GROUP BY aging_bucket.

FR-009 — Tolerance Baseline Change Control

Any change to the 0.05% tolerance baseline (FR-002) requires sign-off from both the Finance Controller and the Compliance/Audit Lead, and must be logged with the prior value, new value, and stated business reason. This prevents either stakeholder from unilaterally loosening or tightening the auto-resolution criteria.

Design rationale: this is the Project B equivalent of Project A's FR-012 — governance over a parameter that, if changed quietly, could either bloat the manual queue (Finance's fear) or hide genuine breaks inside an overly generous auto-resolution rule (Compliance's fear).

FR-010 — Currency Mismatch Handling

A transaction where transaction_id matches but currency does not match between systems is never classified as a Tolerance Match (FR-002), regardless of how small the resulting numeric difference appears after conversion. It is always classified as an Exception (FR-003), since a currency mismatch indicates a data or process error distinct from routine FX rounding.

FR-011 — Bucket-Transition Notification Frequency

A notification per FR-007 is triggered exactly once per Exception — at the moment it first enters the >72-hour bucket. Transitions into the 0–24h and 24–72h buckets (FR-004) do not trigger a notification; they are visible only through the standard aging queue, not pushed. An Exception does not generate repeat >72h notifications for every additional hour it remains unresolved past the 72-hour mark — one notification per exception, at first entry into High priority, is sufficient to satisfy the escalation requirement without creating notification fatigue for the Operations Lead.

Design rationale: this mirrors a real operational tradeoff — over-notifying erodes the signal value of an escalation; a single, well-timed notification at the moment of highest-priority entry is enough to satisfy Compliance's visibility requirement (FR-005/FR-007) without flooding Operations.

3. Out of Scope (FRD-level, consistent with BRD Section 6)

- No SQL query specified above is executed against real or mock data
- No dashboard, report, or visualization is built from FR-007 or FR-008
- No actual internal_ledger, payment_gateway_feed, or clearing_house_file system exists; table/column names above are specification conventions only